

1. Administrative & Study Identification

Full Study Title: A Trial to Evaluate Tolerability and Immunogenicity of V540D in Healthy Adults (V540D-004) **Official Title:** A Phase 1, Randomized, Double-Blind, Comparator-Controlled, Dose-Escalation Trial to Evaluate, Tolerability, and Immunogenicity of V540D in Healthy Adults. **Short Title/Acronym:** V540D-004 Vaccine Study
Protocol Number: V540D-004 **NCT Number:** NCT06688058 **Trial Status:** RECRUITING **Sponsor/Company Name:** Merck **Investigational Product:** V540D (Investigational HPV vaccine) / Active Comparators (Trade Names): GARDASIL®9 (G9), Gardasil, Cervarix (L1 protein, human papillomavirus type 18 vaccine) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of the V540D vaccine in healthy adults.
Secondary Objectives: To evaluate the immunogenicity of V540D compared to GARDASIL®9 (G9) by calculating the Geometric Mean Ratios (GMRs) of specific anti-HPV types.

2.2 Background & Rationale To address the gap in prevention against additional Human Papillomavirus (HPV) types. A standard vaccine to prevent HPV-related cancers is GARDASIL®9 (G9), which protects against 9 types of HPV. However, it does not protect against other oncogenic types. The study vaccine (V540D) is designed to protect against the same HPV types covered by G9, plus additional HPV types, aiming to provide broader protection and further reduce the risk of developing HPV-related cancers later in life.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator-controlled **Blinding:** Double-Blind **Randomization:** Randomized, Dose-Escalation

3.2 Planned Enrollment & Duration Target \$N\$ (Targeted Enrollment): 72 **Trial Status:** Recruiting **Number of Sites:** Multiple U.S. clinical research sites (e.g., Kansas City, MO; Knoxville, TN; Hollywood, FL) **Estimated Study Start:** 20-Dec-2024 **Estimated Primary Completion:** 30-Nov-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy adults. 2. Is in good health based on medical history, physical examination, vital sign (VS) measurements, and electrocardiograms (ECGs) performed before randomization.

4.2 Exclusion Criteria 1. Has a history of abnormal Pap smears, HPV-related external genital lesions (e.g., condyloma acuminata, vulval intraepithelial neoplasia [VIN], or prostatic intraepithelial neoplasia [PIN]) or external genital cancer (e.g., penile cancer), HPV-related vaginal or anal lesions (e.g., condyloma acuminata or vaginal intraepithelial neoplasia [VaIN]) or vaginal or anal cancer. 2. Has a history of cancer (malignancy). 3. Has received any HPV vaccine or is expected to receive any HPV vaccine during the study, outside the protocol.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Randomized to dose-escalated levels of V540D vs. GARDASIL®9. **Route of Administration:** Injection (Vaccine). **Duration of Treatment:** Vaccinations followed by safety and immunogenicity observation periods through November 2026.

5.2 Concomitant & Prohibited Therapy Permitted: Standard routine wellness visits and non-conflicting medications. **Prohibited:** Receipt of any other HPV vaccines outside the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Pre-Randomization Informed Consent, Medical History, Physical Exam, Vital Signs, ECGs
2	Baseline	Day 0 Randomization, First Vaccine Dose Administration, Patient Diary distribution
3	Interim Follow-ups	Scheduled Intervals Safety assessments, Solicited/Unsolicited AE collection, blood draws for immunogenicity (titers)
4	End of Study	Nov 2026 Final Safety Follow-up, ECI (Events of Clinical Interest) review, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: 1. Number of Participants Who Experience a Solicited Injection-Site Adverse Event (AE) (any untoward medical occurrence temporally associated with study intervention). 2. Number of Participants Who Experience a Solicited Systemic AE (including headache, fatigue, nausea, dizziness, muscle aches, joint pain). 3. Number of Participants Who Experience Immediate Reactions Occurring Within 30 Minutes

After Any Vaccination. **Measurement Method:** Clinical investigator assessments and subject diary logs tracking the temporal association of untoward medical occurrences.

7.2 Secondary Outcomes Definition: Total Immunoglobulin G (IgG) Geometric Mean Ratios (GMRs) for anti-HPV types 6, 11, 16, 18, 31, 33, 45, 52, and 58. **Measurement Method:** Central Laboratory Immunogenicity Assay Analysis.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of solicited injection-site and systemic AEs via participant self-reporting/diaries following vaccination; unsolicited AEs captured at clinical visits. **Serious/Medically Attended AEs:** Tracking of MAAEs (visits to ER/urgent care) and Events of Clinical Interest (ECIs) including drug-induced liver injury or potential immune-mediated diseases (pIMDs). **Data Monitoring Committee (DMC):** A safety review committee to oversee dose-escalation milestones and safety thresholds.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all AE/SAE evaluations; Per-Protocol/Immunogenicity Set for evaluating IgG GMRs between groups. **Sample Size Justification:** A targeted enrollment of 72 participants appropriately powered for a Phase 1 safety, tolerability, and dose-escalation evaluation. **Handling of Missing Data:** Standard Phase 1 protocols handling missing participant data or study dropouts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular clinical monitoring of sites for strict adherence to inclusion/exclusion factors and dose-escalation procedures. **Audit Trail:** eCRF locking and compliance with FDA biologics (vaccine) regulations.

11. Study Identifiers & Governance

Primary ID: {{V540D-004}} **Registry Number:** {{NCT06688058}}
Posting ID: {{146486362752953607}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{A Trial to Evaluate Tolerability and Immunogenicity of V540D in Healthy Adults (V540D-004)}} **Official Regulatory Title:** {{A Phase 1, Randomized, Double-Blind, Comparator-Controlled, Dose-Escalation Trial to Evaluate,

Tolerability, and Immunogenicity of V540D in Healthy Adults}}

Trial Status: {{RECRUITING}}

12. Clinical Framework (Vaccine/HPV Prevention Specific)

Primary Condition: {{Healthy / Human Papillomavirus (HPV) Infection Prevention}} **Diagnosis Threshold:** {{Healthy adults with no history of HPV-related external genital lesions or cancers}} **Medical Methodology:** {{Prophylactic Vaccination}} **Optimization Target:** {{Broadened HPV type protection with immunogenicity comparable to GARDASIL®9}}

13. Study Procedures & Methodology

Management Pathway: {{Phase 1 Vaccine Dose-Escalation & Immunogenicity Tracking}} **Education Protocol (HCP):** {{Training on dose escalation rules, blinded administration, and specific safety monitoring}} **Education Protocol (Patient):** {{Instructions for capturing solicited AEs (e.g., headache, injection site pain, fatigue) via diary}} **Quality Audit Mechanism:** {{ECG review, vital sign assessments, and tracking of immediate reactions within 30 minutes of vaccination}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to Study Completion (estimated 2 years)}} **Onsite Visit Cadence:** {{Scheduled vaccination administration and post-vaccination safety visits}} **Remote Monitoring Frequency:** {{Patient diary logging for solicited AEs in days immediately following vaccination}} **Checklist Review Interval:** {{Continual review of immediate reactions, unsolicited AEs, and ECIs at follow-up clinical visits}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				